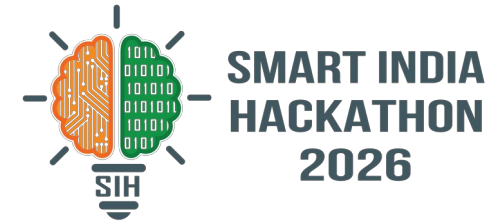


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26173
- **Problem Statement Title**- iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 188881
- **Team Name (Registered on portal)** Team Turtle



Website



Demo film



Code

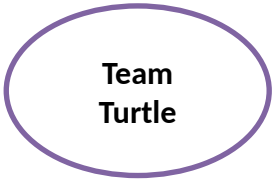


APK

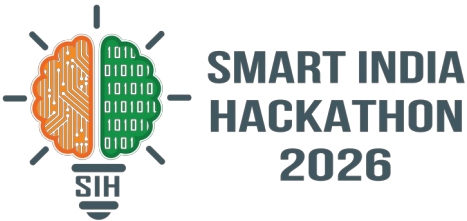


iTantra

Speak. Send text. Hear it back — offline.



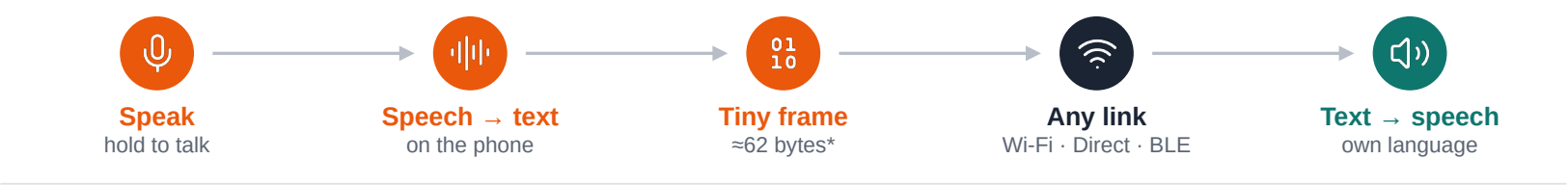
iTantra: Speak. Send Text. Hear It Back — Offline



Proposed Solution (Describe your Idea/Solution/Prototype)

Problem: when towers fail after a flood or cyclone, only thin links remain — Bluetooth, a crowded hotspot or LoRa radio at ~250 bps. Voice needs 256 kbps raw, ~12 kbps with a good codec.

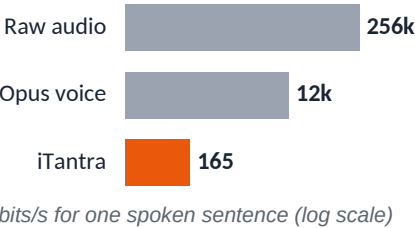
Our idea: send the meaning, not the sound. Speech becomes text on the phone, only the text travels, and it is spoken again on arrival. **No tower, no internet, no cloud.**



Detailed explanation of the proposed solution

- 10 Indian languages.** Hindi, Bengali, Marathi, Telugu, Tamil, Gujarati, Kannada, Malayalam, Odia, English — speech in and out.
- ALERT mode.** Plays on the alarm stream at full volume with vibration (16/16 in test).
- SOS.** One tap sends 1 of 6 templates with GPS/NavIC location, read aloud on arrival.
- Talk your way.** Push-to-talk, hands-free call mode and live captions.

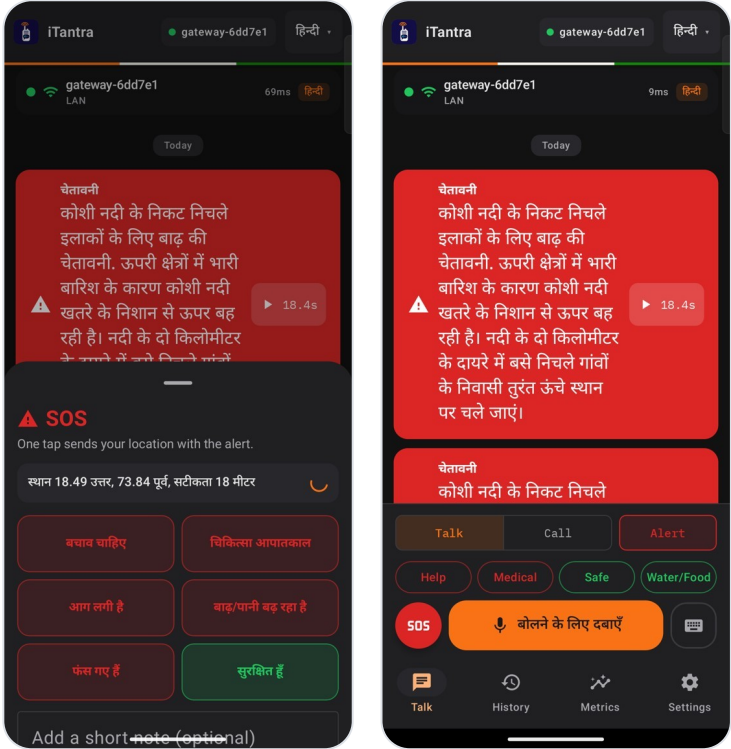
How it addresses the problem



- ≈1,500× less data** than raw audio — a sentence fits LoRa's ~250 bps.
- 62 B** per sentence in binary vs 524 B as JSON.
- Still speech at the end**, so people who cannot read are included.

Innovation and uniqueness of the solution

- Speech is the codec.** Text on the wire; the voice is rebuilt in the listener's language.
- One frame, any link.** Wi-Fi, then Wi-Fi Direct (4 s), then Bluetooth LE (8 s), with no user action.
- Trusted alerts.** ALERTs are Ed25519-signed; messages are encrypted after QR pairing.
- Adapts to the phone.** LITE profile turns on by itself below 6 GB RAM.



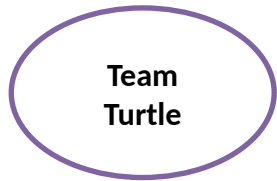
Sender · SOS

Receiver · ALERT

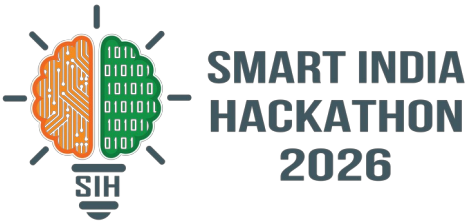


Real screens from the running prototype (v0.1.0), vivo I2202, Snapdragon 870. [Try it: itantra-106.pages.dev](#)

*Median of 310 FLEURS sentences (unit test), before encryption and link headers. Bars assume a ~3 s sentence.



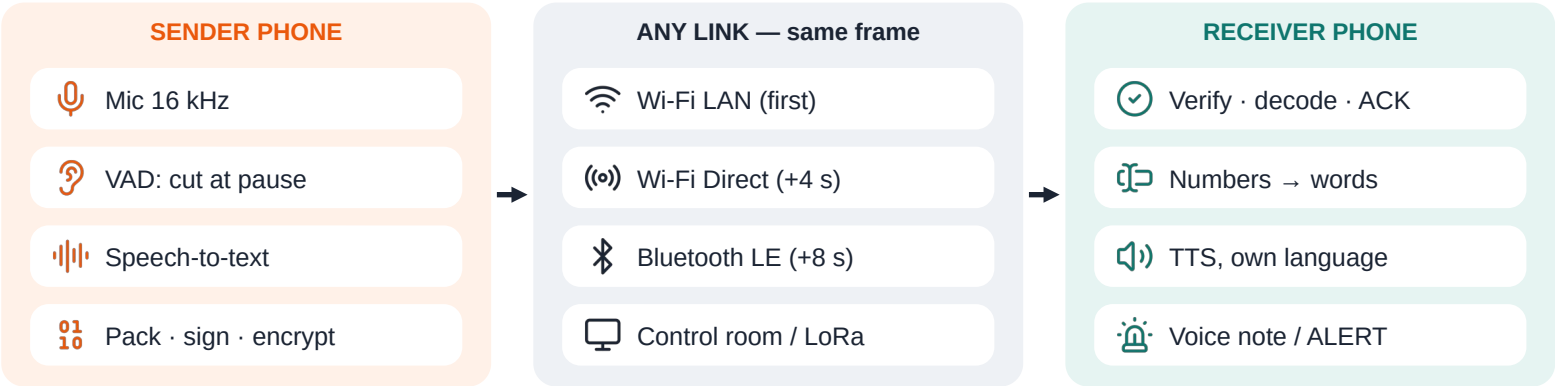
TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

Layer	Technology
App	Kotlin 2.4, Jetpack Compose; Android 8.0+, 64-bit ARM
Voice detection	Silero VAD (0.6 MB) — a pause ends the sentence
Speech-to-text	SraVaani-1.0 (IISc), int8 ONNX on sherpa-onnx
Text-to-speech	VITS Rasa (6 languages), Indic-Mio via llama.cpp (4)
Links	Wi-Fi mDNS/TCP, Wi-Fi Direct, BLE GATT
Security	X25519, ChaCha20-Poly1305, Ed25519; QR pairing
Radio	ESP32-S3 + SX1262 LoRa, 865–867 MHz, RadioLib
Control room	Python gateway, SQLite log, CAP 1.2 alert parser

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Built and running

Push-to-talk, call mode, 10-language STT/TTS, ALERT, SOS, LITE profile, control room.

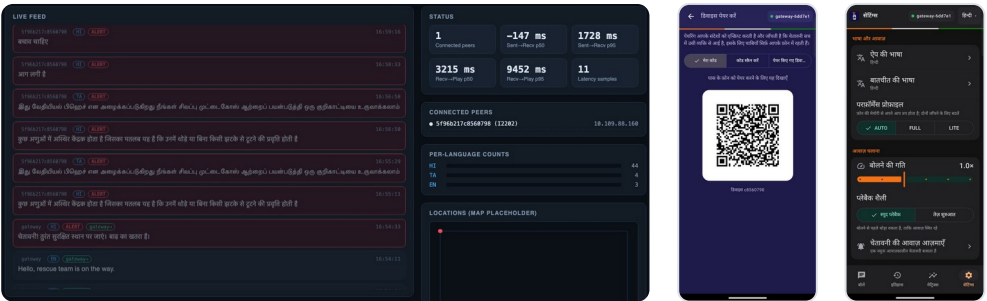
Partial · test pending

Encryption covers binary mode; LoRa firmware compiles; Wi-Fi Direct/BLE two-phone test.

Planned

On-device translation (IndicTrans2), smaller voices, phone-to-phone mesh relay.

Working prototype — real screens



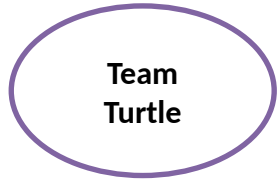
PC control room

QR pairing

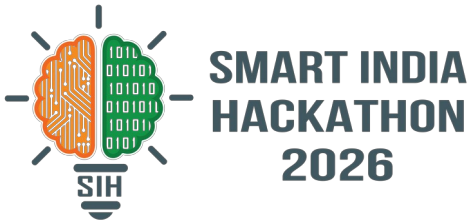
LITE profile

- Prototype v0.1.0, APK 60 MB
- Models 1.2 GB, installed once
- Tested on Snapdragon 870 and 8 Gen 3

[Reviewer install guide](#)



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Measured: 4 GB-class memory test, in-app logging

14×

faster than speech

real-time factor 0.07, 10/10 languages

0.6 s

speech end → text

median; range 0.42–0.70 s

765 MB

app RAM when idle

1.8 GB peak while speaking

0.3%

idle CPU

of one core, push-to-talk

Alive

app not killed

Android closed 115 other processes

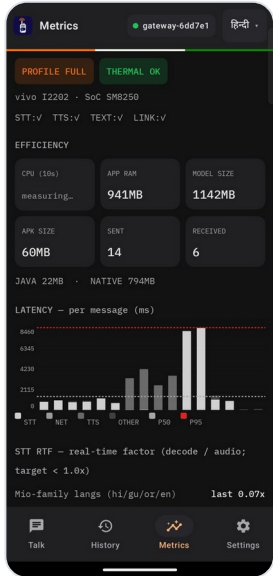
Method: 12 GB vivo I2202 (Snapdragon 870) with RAM reserved so only ~1.7 GB stayed free, LITE profile; the chip was not slowed, so a budget phone will be slower. Also measured: 41 ms message → ACK over Wi-Fi · 0 audio underruns in 5 languages · word error rate 5.9% (preliminary, 10 clips).

Potential challenges and risks	Strategies for overcoming these challenges
Voice starts 3–6 s after arrival in LITE	Keep the listener's language loaded; add small, fast voices for hi/gu/or/en
1.14 GB of speech models on the phone	Download only the languages a user needs; installer checks every file's SHA-256
Wi-Fi Direct / BLE: two-phone test pending	Field tests on two phones; Wi-Fi stays the primary link; LoRa bridge for range
Noise, accents and code-mixing	Boost alert keywords; 30-clip-per-language evaluation; typed text as a fallback

Next steps: two-phone field test of every link · flash and range-test the LoRa bridge · 30-clip-per-language accuracy run · smaller voices for hi/gu/or/en.

Viability

- ₹0 per message: no airtime, SIM, server or cloud fees.
- Phones people own: Android 8.0+; LITE mode for less RAM.
- Low-cost range: ESP32-S3 + LoRa node ≈ ₹1,900–3,200.
- Open licences: MIT, Apache-2.0 and CC-BY-4.0 models.
- Fits alert systems: the control room reads CAP 1.2 alert files.



In-app metrics (real screen)

Potential impact on the target audience



Disaster response teams

Coordinate rescue by voice when towers and internet are down.



Coastal fishing communities

Cyclone warnings spoken in their own language, with no data plan.



Low-literacy users

An alert they can hear instead of read, in 10 languages.



Remote field staff

A voice link for teams working where coverage is poor.

≈1,500×

less data than raw audio*

1.03 s

ALERT arrival → spoken aloud (Wi-Fi)

10

Indian languages, speech in and out

Benefits of the solution (social, economic, environmental, etc.)



Social

- Alerts spoken aloud in Indian languages — people who cannot read are included.
- SOS carries location to rescuers in one tap.



Economic

- No airtime, SIM, cloud or licence fees.
- Runs on phones people own; a LoRa node costs ≈ ₹1,900–3,200.



Environmental

- Fewer bytes per message means less radio time on air.
- No servers or data centres needed. Battery gain not yet measured.

Fits India's alert system: the control room reads CAP 1.2 alert files — the format behind [NDMA SACHET](#) — and re-broadcasts them as spoken alerts in each listener's language.

*62 B per ~3 s sentence (unit-test median) vs 256 kbps raw audio. No field pilot yet.

Photos, Wikimedia Commons: [NDRF – MHA](#), [GODL-India](#) · [fishermen – Saral Shots](#), CC0 · [woman on phone – Ammalu](#), CC BY-SA 3.0 · [rail workers – David Brossard](#), CC BY-SA 2.0

Details / Links of the reference and research work

Open models

[SraVaani-1.0 STT · ARTPARK, IISc](https://huggingface.co/ARTPARK-IISc/SraVaani-1.0)
huggingface.co/ARTPARK-IISc/SraVaani-1.0

[Indic-Mio TTS · SPRING Lab, IIT-M](https://huggingface.co/SPRINGLab/Indic-Mio)
huggingface.co/SPRINGLab/Indic-Mio

[VITS Rasa 13 TTS · AI4Bharat](https://huggingface.co/ai4bharat/vits_rasa_13)
huggingface.co/ai4bharat/vits_rasa_13

[Silero VAD](https://github.com/snakers4/silero-vad)
github.com/snakers4/silero-vad

[IndicTrans2 \(planned translation\) · AI4Bharat](https://github.com/AI4Bharat/IndicTrans2)
github.com/AI4Bharat/IndicTrans2

Runtimes and libraries

[sherpa-onnx](https://github.com/k2-fsa/sherpa-onnx)
github.com/k2-fsa/sherpa-onnx

[llama.cpp / ggml](https://github.com/ggml-org/llama.cpp)
github.com/ggml-org/llama.cpp

[mio-tts-cpp](https://github.com/mmmnga/mio-tts-cpp)
github.com/mmmnga/mio-tts-cpp

[Bouncy Castle \(encryption\)](https://www.bouncycastle.org)
www.bouncycastle.org

[RadioLib \(LoRa\)](https://github.com/jgromes/RadioLib)
github.com/jgromes/RadioLib

Standards and data

[OASIS CAP 1.2 alerts](https://docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html)
docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html

[NDMA SACHET](https://sachet.ndma.gov.in)
sachet.ndma.gov.in

[Google FLEURS \(STT test set\)](https://huggingface.co/datasets/google/fleurs)
huggingface.co/datasets/google/fleurs

[Codec2 \(bitrate baseline\)](https://github.com/drowe67/codec2)
github.com/drowe67/codec2

[Meshtastic \(LoRa mesh\)](https://meshtastic.org)
meshtastic.org



[Website](#)
overview · results



[Demo film](#)
1 min 55 s



[Code](#)
GitHub



[APK](#)
Android 8+

Android guides

[Wi-Fi Direct](https://developer.android.com/develop/connectivity/wifi/wifi-direct)
developer.android.com/develop/connectivity/wifi/wifi-direct

[Bluetooth and BLE](https://developer.android.com/develop/connectivity/bluetooth)
developer.android.com/develop/connectivity/bluetooth

[Audio focus \(ALERT\)](https://developer.android.com/media/optimize/audio-focus)
developer.android.com/media/optimize/audio-focus

Our work

[Source code](https://github.com/prasadxa/itantra)
github.com/prasadxa/itantra

[4 GB-class memory test](https://github.com/prasadxa/itantra/tree/main/tools/lowram)
github.com/prasadxa/itantra/tree/main/tools/lowram

[Model pack \(1.2 GB\)](https://r2.dev/models/v1/manifest.json)
r2.dev/models/v1/manifest.json

[Website and results](https://itantra-106.pages.dev)
itantra-106.pages.dev

[Demo film, 1 min 55 s](https://itantra-106.pages.dev/demo)
itantra-106.pages.dev/demo

[Reviewer install guide](https://itantra-106.pages.dev/#try)
itantra-106.pages.dev/#try